

National Health and Nutrition Examination Survey

2017-2018 Data Documentation, Codebook, and Frequencies

Blood Pressure (BPX_J)

Data File: BPX_J.xpt

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Component Description

This section provides data for three consecutive blood pressure (BP) measurements and other methodological measurements to obtain an accurate BP. Heart rate or pulse, depending on age, are also reported.

Eligible Sample

BP is measured on participants 8 years and older; radial (or brachial, if necessary) pulse is measured on participants 8 years and older; and heart rate is measured on children 0-7 years of age.

Protocol and Procedure

After resting quietly in a seated position for 5 minutes and after the participant's maximum inflation level (MIL) has been determined, three consecutive BP readings are obtained. If a BP measurement is interrupted or incomplete, a fourth attempt may be made. All BP determinations (systolic and diastolic) are taken in the mobile examination center (MEC). Participants with any of the following on both arms are excluded from the exam: rashes, gauze dressings, casts, edema, paralysis, tubes, open sores or wounds, withered arms, a-v shunts, and radical mastectomy. BP measurements are taken in the right arm unless specific conditions prohibit the use of the right arm, or, if participants report any reason that the BP measurements should not be taken in the right arm. Prior to BP measurements, upper arm circumference is measured, which is done to guide selection of cuff size. Details on the protocol for obtaining upper arm circumference is described in the [Physician Examination Procedures Manual](#). The following criteria were used to guide selection of cuff size:

Cuff size	Bladder width (cm)	Bladder length (cm)	Arm circumference
Child	9	17	17.0-21.9
Adult	12	22	22.0-29.9
Large adult	15	32	30.0-37.9
Adult thigh	18	35	38.0-47.9

Quality Assurance & Quality Control

The BP examiners are certified for BP measurement through a training program from Shared Care Research and Education Consulting. Certification is achieved when physician examiners meet all requirements of the training program. The initial training includes:

- Didactic instruction about BP measurement and propensity for error;
- Listening to systolic and diastolic BP sounds using a standardized audio-video presentation;

- An audio-video test;
- A written exam; and
- Simultaneously listening to BP on a minimum of 20 volunteers, of all ages, through a Y-stethoscope to compare measures with a gold standard instructor who does the same.

Certification requires a score of 100% on the audio-video test. Correct answers on the audio-video test require the examiner's answers and the standardized test answers to be within +/- 2 mmHg (millimeter of mercury) on 92% of 24 measures (12 systolic, 12 diastolic), and within +/- 4 mmHg on 100% of 24 measures (12 systolic, 12 diastolic).

For more details on the procedures for measuring heart rate, pulse, and BP and for details related to quality assurance/quality control for this component, please refer to the [Physician Examination Procedures Manual](#).

Data Processing and Editing

The following are some specifications used in capturing the BP data:

- Systolic BP and maximum inflation level cannot be greater than 300 mmHg;
- Systolic and diastolic BP measurements and the maximum inflation level can be even numbers only;
- Systolic blood pressure must be greater than diastolic BP;
- If there is no systolic BP, there can be no diastolic BP. (There can be a systolic measurement without a diastolic measurement.); and
- Diastolic BP can be zero.

The physicians attempting to obtain BP may enter comments related to reasons for incomplete data. This data is recorded in the BPX dataset as PEASCCT1. For this variable PEASCCT1=3, or "time constraints" was assigned if the recorded comment was either "no time" or "came late/left early." PEASCCT1=4, or "other", was assigned based on a variety of comments, including participant ill/emergency. There were no other recodes to this variable prior to data release.

Analytic Notes

Exam sample weights should be used for analyses. Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 0 YEARS - 150 YEARS

PEASCCT1 - Blood Pressure Comment

Variable Name: PEASCCT1
SAS Label: Blood Pressure Comment
English Text: Blood Pressure Comment
Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Safety exclusion	0	0	
2	SP refusal	34	34	
3	Time constraints	317	351	
4	Other	93	444	
.	Missing	8260	8704	

BPXCHR - 60 sec HR (30 sec HR * 2)**Variable Name:** BPXCHR**SAS Label:** 60 sec HR (30 sec HR * 2)**English Text:** 60 sec HR (30 sec HR * 2)**Target:** Both males and females 0 YEARS - 7 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
60 to 180	Range of Values	1539	1539	
.	Missing	7165	8704	

BPAARM - Arm selected

Variable Name: BPAARM

SAS Label: Arm selected

English Text: Arm selected:

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Right	6784	6784	
2	Left	33	6817	
8	Could not obtain	0	6817	
.	Missing	1887	8704	

BPACSZ - Coded cuff size

Variable Name: BPACSZ

SAS Label: Coded cuff size

English Text: Cuff size (cm) (width X length)

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Infant (6X12)	0	0	
2	Child (9X17)	408	408	
3	Adult (12X22)	2245	2653	
4	Large (15X32)	3123	5776	
5	Thigh (18X35)	1036	6812	
.	Missing	1892	8704	

BPXPLS - 60 sec. pulse (30 sec. pulse * 2)**Variable Name:** BPXPLS**SAS Label:** 60 sec. pulse (30 sec. pulse * 2)**English Text:** 60 sec. pulse (30 sec. pulse * 2)**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
34 to 136	Range of Values	6742	6742	
.	Missing	1962	8704	

BPXPULS - Pulse regular or irregular?

Variable Name: BPXPULS

SAS Label: Pulse regular or irregular?

English Text: Pulse regular or irregular?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Regular	8054	8054	
2	Irregular	227	8281	
.	Missing	423	8704	

BPXPTY - Pulse type

Variable Name: BPXPTY

SAS Label: Pulse type

English Text: Pulse type

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Radial	6706	6706	
2	Brachial	36	6742	
8	Could not obtain	0	6742	
.	Missing	1962	8704	

BPXML1 - MIL: maximum inflation levels (mm Hg)

Variable Name: BPXML1
SAS Label: MIL: maximum inflation levels (mm Hg)
English Text: MIL: maximum inflation levels (mm Hg)
Target: Both males and females 8 YEARS - 150 YEARS
Hard Edits: 70.0000 to 300.0000

Code or Value	Value Description	Count	Cumulative	Skip to Item
100 to 260	Range of Values	6731	6731	
.	Missing	1973	8704	

BPXSY1 - Systolic: Blood pres (1st rdg) mm Hg**Variable Name:** BPXSY1**SAS Label:** Systolic: Blood pres (1st rdg) mm Hg**English Text:** Systolic: Blood pressure (first reading) mm Hg**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
72 to 228	Range of Values	6302	6302	
.	Missing	2402	8704	

BPXDI1 - Diastolic: Blood pres (1st rdg) mm Hg**Variable Name:** BPXDI1**SAS Label:** Diastolic: Blood pres (1st rdg) mm Hg**English Text:** Diastolic: Blood pressure (first reading) mm Hg**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 136	Range of Values	6302	6302	
.	Missing	2402	8704	

BPAEN1 - Enhancement used first reading

Variable Name: BPAEN1

SAS Label: Enhancement used first reading

English Text: Enhancement used first reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	2	2	
2	No	6300	6302	
8	Could not obtain	0	6302	
.	Missing	2402	8704	

BPXSY2 - Systolic: Blood pres (2nd rdg) mm Hg**Variable Name:** BPXSY2**SAS Label:** Systolic: Blood pres (2nd rdg) mm Hg**English Text:** Systolic: Blood pressure (second reading) mm Hg**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
72 to 236	Range of Values	6563	6563	
.	Missing	2141	8704	

BPXDI2 - Diastolic: Blood pres (2nd rdg) mm Hg**Variable Name:** BPXDI2**SAS Label:** Diastolic: Blood pres (2nd rdg) mm Hg**English Text:** Diastolic: Blood pressure (second reading) mm Hg**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 136	Range of Values	6563	6563	
.	Missing	2141	8704	

BPAEN2 - Enhancement used second reading

Variable Name: BPAEN2

SAS Label: Enhancement used second reading

English Text: Enhancement used second reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	177	177	
2	No	6386	6563	
8	Could not obtain	0	6563	
.	Missing	2141	8704	

BPXSY3 - Systolic: Blood pres (3rd rdg) mm Hg**Variable Name:** BPXSY3**SAS Label:** Systolic: Blood pres (3rd rdg) mm Hg**English Text:** Systolic: Blood pressure (third reading) mm Hg**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
72 to 238	Range of Values	6538	6538	
.	Missing	2166	8704	

BPXDI3 - Diastolic: Blood pres (3rd rdg) mm Hg**Variable Name:** BPXDI3**SAS Label:** Diastolic: Blood pres (3rd rdg) mm Hg**English Text:** Diastolic: Blood pressure (third reading) mm Hg**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 134	Range of Values	6538	6538	
.	Missing	2166	8704	

BPAEN3 - Enhancement used third reading

Variable Name: BPAEN3

SAS Label: Enhancement used third reading

English Text: Enhancement used third reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	208	208	
2	No	6330	6538	
8	Could not obtain	0	6538	
.	Missing	2166	8704	

BPXSY4 - Systolic: Blood pres (4th rdg) mm Hg**Variable Name:** BPXSY4**SAS Label:** Systolic: Blood pres (4th rdg) mm Hg**English Text:** Systolic: Blood pressure (fourth reading if necessary) mm Hg**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
74 to 234	Range of Values	556	556	
.	Missing	8148	8704	

BPXDI4 - Diastolic: Blood pres (4th rdg) mm Hg**Variable Name:** BPXDI4**SAS Label:** Diastolic: Blood pres (4th rdg) mm Hg**English Text:** Diastolic: Blood pressure (fourth reading if necessary) mm Hg**Target:** Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 118	Range of Values	556	556	
.	Missing	8148	8704	

BPAEN4 - Enhancement used fourth reading

Variable Name: BPAEN4

SAS Label: Enhancement used fourth reading

English Text: Enhancement used fourth reading

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	129	129	
2	No	427	556	
8	Could not obtain	0	556	
.	Missing	8148	8704	